VII SEMESTER (PROFESSIONAL ELECTIVE-III)	L	T	P	C
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20CS7E01 - CRYPTOGRAPHY AND NETWORK SECURITY				

COURSE OBJECTIVES:

The main objectives of this course are to explore the working principles and utilities of various crypto-graphic algorithms including secret key cryptography, hashes and message digests, public key algorithms, design issues and working principles of various authentication protocols and various secure communication standards including Kerberos, IPsec, and SSL/TLS.

COURSE OUTCOMES:

At the end of the course, student will be able to

CO1: Explain different security threats and countermeasures and foundation course of cryptography mathematics.

CO2: Classify the basic principles of symmetric key algorithms and operations of some symmetric key algorithms and asymmetric key cryptography

CO3: Revise the basic principles of Public key algorithms and Working operations of some Asymmetric key algorithms such as RSA, ECC and some more

CO4: Design applications of hash algorithms, digital signatures and key management techniques

CO5: Determine the knowledge of Application layer, Transport layer and Network layer security Protocols such as PGP, S/MIME, SSL,TSL, and IPsec.

UNIT -I:

Basic Principles: Security Goals, Security Attacks,Security Services,Security Mechanisms, Symmetric Cipher Model,Substitution Techniques,Transposition Technique, Phishing and Defensive Measure,Web-Based Attacks,Structured Query Language(SQL) Injection attacks.

UNIT -II:

Traditional Block Cipher Structure:Stream Cipher and Block Cipher.

Symmetric Encryption: Mathematics of Symmetric Key Cryptography, Introduction to Modern Symmetric Key Ciphers, Data Encryption Standard,IDEA(International Data Encryption Algorithm),Advanced Encryption Standard.

UNIT -III:

Asymmetric Encryption: Mathematics of Asymmetric Key Cryptography,Asymmetric Key Cryptography, RSA Algorithm,Algorithm for Diffe-Hellman Key Exchange,Elliptic Curve Cryptography.

UNIT -IV:

Data Integrity, Digital Signature Schemes & Key Management:Hash Function, Applications of Cryptographic Hash Functions,SHA(Secure Hash Algorithm),Message Integrity and Message Authentication, , Digital Signature, Key Management and Distribution.

UNIT -V:

Network Security-I: Remote User Authentication Principles,Kerberos,Web Security ,Security at application layer: PGP and S/MIME, Security at the Transport Layer: SSL and TLS, Network Security-II: Secure Shell(SSH),Security at the Network Layer: IPSec, System Security

Text Books:

1. Cryptography and Network Security, 3rd Edition Behrouz A Forouzan, Deb deep Mukhopadhyay, McGraw Hill,2015
2. Cryptography and Network Security,4th Edition, William Stallings, (6e) Pearson,2006
3. Everyday Cryptography, 1st Edition, Keith M.Martin, Oxford,2016

Reference Books:

1. Network Security and Cryptography, 1st Edition, Bernard Meneges, Cengage Learning,2018.